LIBRARY MANAGEMENT PLATFORM

Mini Project Report

Unit 1 – Product Engineering & Design Thinking

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1.Introduction

The Library Management Platform is a proposed software product designed to improve and simplify library operations. It focuses on important activities such as searching books, borrowing books, returning books, managing fines, and administration.

The platform aims to reduce manual administrative work and provide a better digital catalog experience for students, librarians, and administrators. It follows a product engineering and design thinking approach to understand user needs and develop a useful and efficient library management solution.

The proposed platform can help make library operations more organized, automated, and easier to manage.

2. Problem Statement

Traditional library management involves several manual activities such as maintaining book records, searching for books, recording borrowing and returning details, and tracking overdue books and fines.

These manual processes can make library operations time-consuming and may lead to difficulties in maintaining accurate records. Users may also face difficulty in quickly finding book availability.

The proposed Library Management Platform aims to solve these problems by providing a centralized and automated system for managing books, users, circulation, fines, and notifications. This can make library operations more organized, efficient, and easier to manage.

3. Objectives

The main objectives of the Library Management Platform are:

1. To simplify library book searching and catalog management.

2. To manage book borrowing and returning in an organized way.

3. To automate fine calculation for overdue books.

4. To provide role-based access for students, faculty, librarians, and administrators.

5. To provide notifications for important library activities.

6. To reduce manual administrative work and improve the efficiency of library operations.

7. To provide basic reports and information for library administrators.

4. Users of the System

The Library Management Platform is designed for different types of users. Each user has different responsibilities and requirements.

User | Role and Requirements

Student | Searches for books, checks availability, borrows and returns books, and receives notifications.

Librarian | Manages books, memberships, borrowing and returning, fines, and library records.

Administrator | Monitors library operations, manages users, and views reports and analytics.

5. Design Thinking

Design Thinking is a user-centered approach used to understand users' problems and develop suitable solutions. It helps the project team focus on the actual needs of students, librarians, and administrators.

The Library Management Platform follows five main stages of Design Thinking:

1. Empathize – Understand the needs, problems, and experiences of library users.

2. Define – Clearly identify and define the main problems faced by users.

3. Ideate – Generate different ideas and possible solutions to the identified problems.

4. Prototype – Create a basic representation of the proposed solution.

5. Test – Evaluate the solution with users and collect feedback for improvement.

Application to Library Management

The Design Thinking approach can be applied to the Library Management Platform as follows:

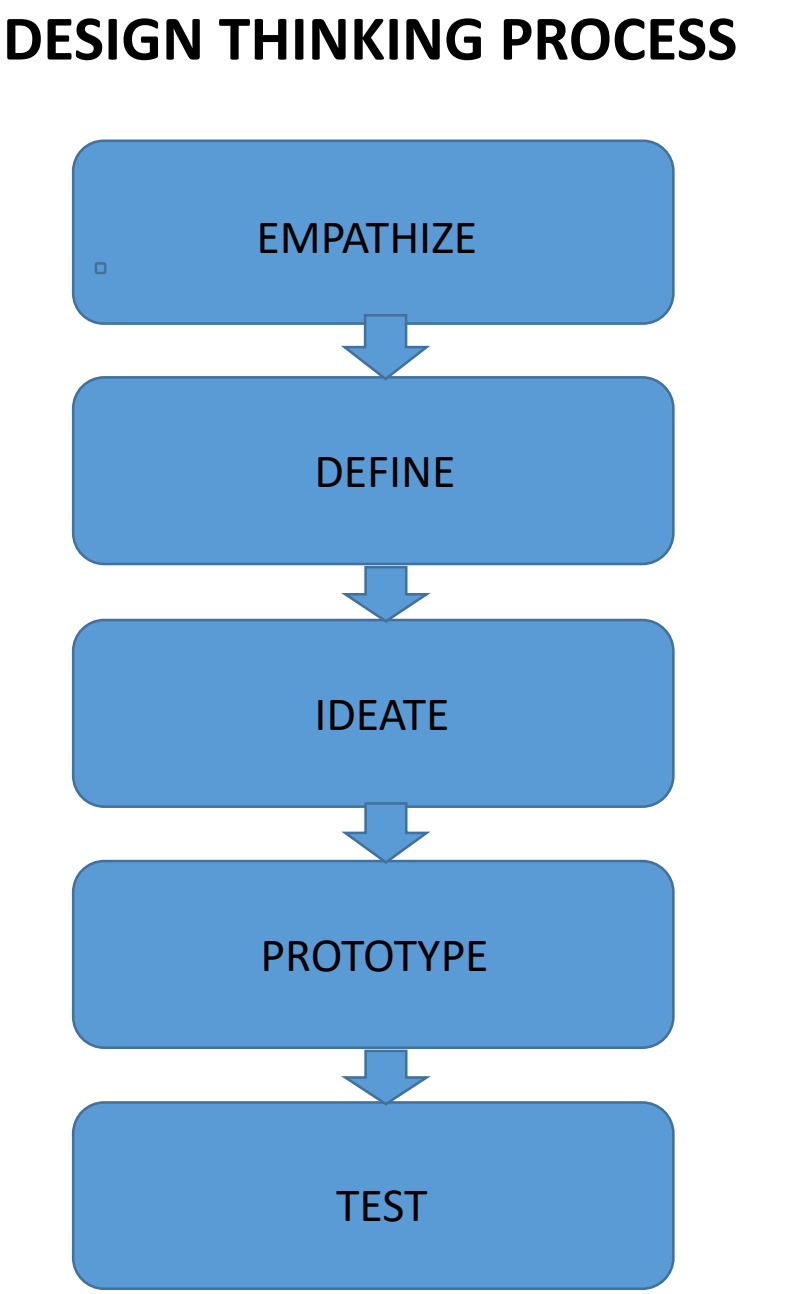
- Empathize: Understand difficulties faced by students and librarians.

- Define: Identify problems such as difficulty finding books and manually tracking overdue books.

- Ideate: Develop ideas such as automated notifications, QR-based membership, and live book availability.

- Prototype: Prepare a basic design of the library platform and its main screens.

- Test: Collect feedback from users and improve the proposed design.



6.Requirements

6.1 Functional Requirements

Functional requirements describe the main functions that the Library Management Platform should provide.

Now create a table with 3 columns and 6 rows.

ID

Requirement

Description

FR-01

Authentication & RBAC

Provide login and role-based access for Admin, Librarian, Student, and Faculty.

FR-02

Book Catalog

Allow authorized users to add, view, update, and delete book records.

FR-03

Circulation

Manage book issue, return, and due-date information.

FR-04

Fine Engine

Automatically calculate fines for overdue books.

FR-05

Notification

Send notifications related to due dates, overdue books, and other library activities.

6.2 Non-Functional Requirements.

Non-functional requirements describe the quality, performance, security, and maintainability expected from the platform.

Create another 3-column table with 5 rows:

ID	Requirement	Description
NFR-01	Security	Use secure authentication, password hashing, JWT, and HTTPS.
NFR-02	Performance	Book search API should respond within 500 ms under concurrent load.
NFR-03	Availability & Scalability	Target 99.9% availability and support horizontal scaling.
NFR-04	Maintainability	The system should be easy to maintain using Infrastructure as Code.

7. MVP Features

MVP stands for Minimum Viable Product. It represents the basic version of the Library Management Platform that contains the most important features required for initial use.

The proposed MVP of the Library Management Platform includes:

1. User Login and Profile – Allows users to log in and access features according to their roles.

2. Role-Based Access Control – Provides different access levels for students, faculty, librarians, and administrators.

3. Book Search and Availability – Allows users to search for books and check their availability.

4. Book Issue and Return – Supports basic circulation operations such as issuing and returning books.

5. Automated Fine Calculation – Calculates fines for overdue books automatically.

6. Basic Admin Reporting – Provides basic information and reports for library administrators.

Feature	**Main User**

Login & Profile	All Users
Book Search	Student / Faculty
Book Issue & Return	Student / Faculty / Librarian
Fine Calculation	Librarian / Administrator
Admin Reports	Administrator

8. System Architecture

The proposed Library Management Platform follows a layered architecture. Different components work together to provide library management services.

The main components of the proposed system are:

1. Users – Students, Faculty, Librarians, and Administrators interact with the platform.

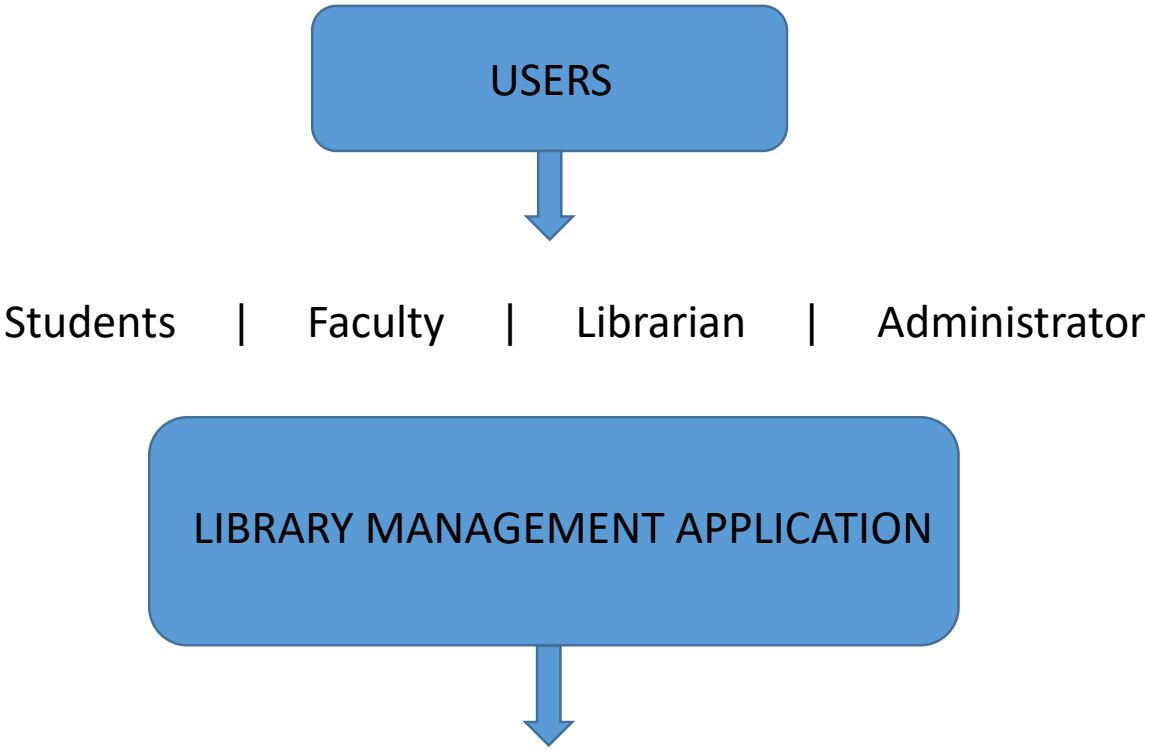
2. Application Layer – Handles user requests and provides library services such as authentication, book catalog management, circulation, fines, and notifications.

3. Database – Stores information about users, books, borrowing records, returns, and fines.

4. Notification System – Sends notifications related to due dates, overdue books, and other library activities.

5. Administration and Monitoring – Supports administration, reporting, and monitoring of the platform.

SYSTEM ARCHITECTURE – LIBRARY MANAGEMENT PLATFORM



9. Project Roadmap

The Library Management Platform is planned in multiple phases. Each phase focuses on a specific part of product engineering, from understanding the library domain to deployment and monitoring.

The proposed roadmap consists of the following phases:

Phase 1 – Domain Analysis

Understand library operations, users, problems, and requirements.

Phase 2 – Design Thinking

Understand user needs and develop suitable solutions using the Design Thinking process.

Phase 3 – Agile Sprint Planning

Plan the project work into smaller tasks and development sprints.

Phase 4 – Core Application Development

Develop the main library management features such as catalog management, circulation, fines, and notifications.

Phase 5 – DevOps Automation

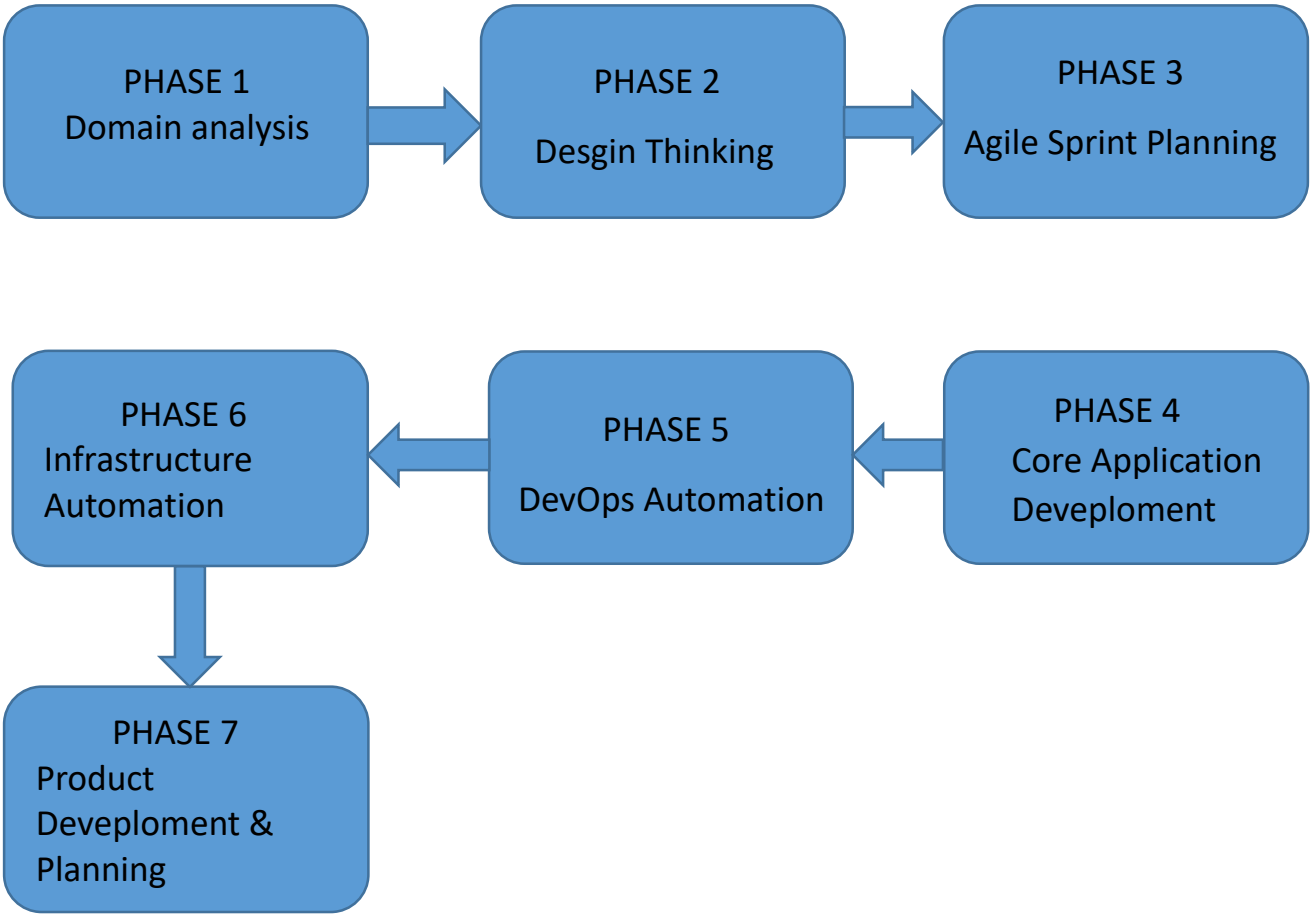
Introduce CI/CD, automated testing, and containerization.

Phase 6 – Infrastructure Automation

Use infrastructure automation tools such as Terraform and Ansible.

Phase 7 – Production Deployment and Monitoring

Deploy the platform and monitor it using technologies such as Kubernetes, Prometheus, and Grafana.



10. Expected Benefits

The proposed Library Management Platform is expected to provide the following benefits:

1. Faster Book Searching – Users can search the digital book catalog more easily.

2. Better Record Management – Book, user, borrowing, and fine records can be maintained in a centralized system.

3. Reduced Manual Work – Automation can reduce repetitive administrative activities.

4. Improved Circulation Management – Book issue and return activities can be managed in an organized way.

5. Automated Fine Management – Overdue fines can be calculated automatically.

6. Timely Notifications – Users can receive reminders and notifications related to library activities.

7. Better Administration – Administrators can access basic reports and information about library operations.

8. Improved User Experience – A digital platform can make library services easier and more convenient for users.

11. Future Scope

The Library Management Platform can be further improved in future by adding more advanced features and capabilities.

Possible future improvements include:

1. Mobile Application – Provide library services through a mobile application.

2. Self-Checkout – Allow users to perform book checkout with less manual assistance.

3. QR-Based Membership – Use QR codes for digital membership cards.

4. Live Inventory Dashboard – Provide real-time information about book availability and inventory.

5. Advanced Notifications – Provide automated email and SMS reminders for due dates and overdue books.

6. Advanced Reporting and Analytics – Provide more detailed information and analytics for administrators.

7. Cloud-Based Scaling and Monitoring – Improve the platform's deployment, scalability, and monitoring capabilities.

12. Conclusion

The Library Management Platform is a proposed solution for improving and automating traditional library operations. The project applies product engineering and Design Thinking principles to understand user needs and identify suitable solutions.

The proposed platform focuses on important activities such as book searching, catalog management, book issue and return, fine calculation, notifications, and administration. It can help reduce manual work and make library operations more organized and efficient.

The project also provides a structured roadmap covering domain analysis, Design Thinking, application development, DevOps automation, infrastructure automation, and production monitoring. Overall, the proposed platform provides a foundation for developing a modern and user-focused library management solution.